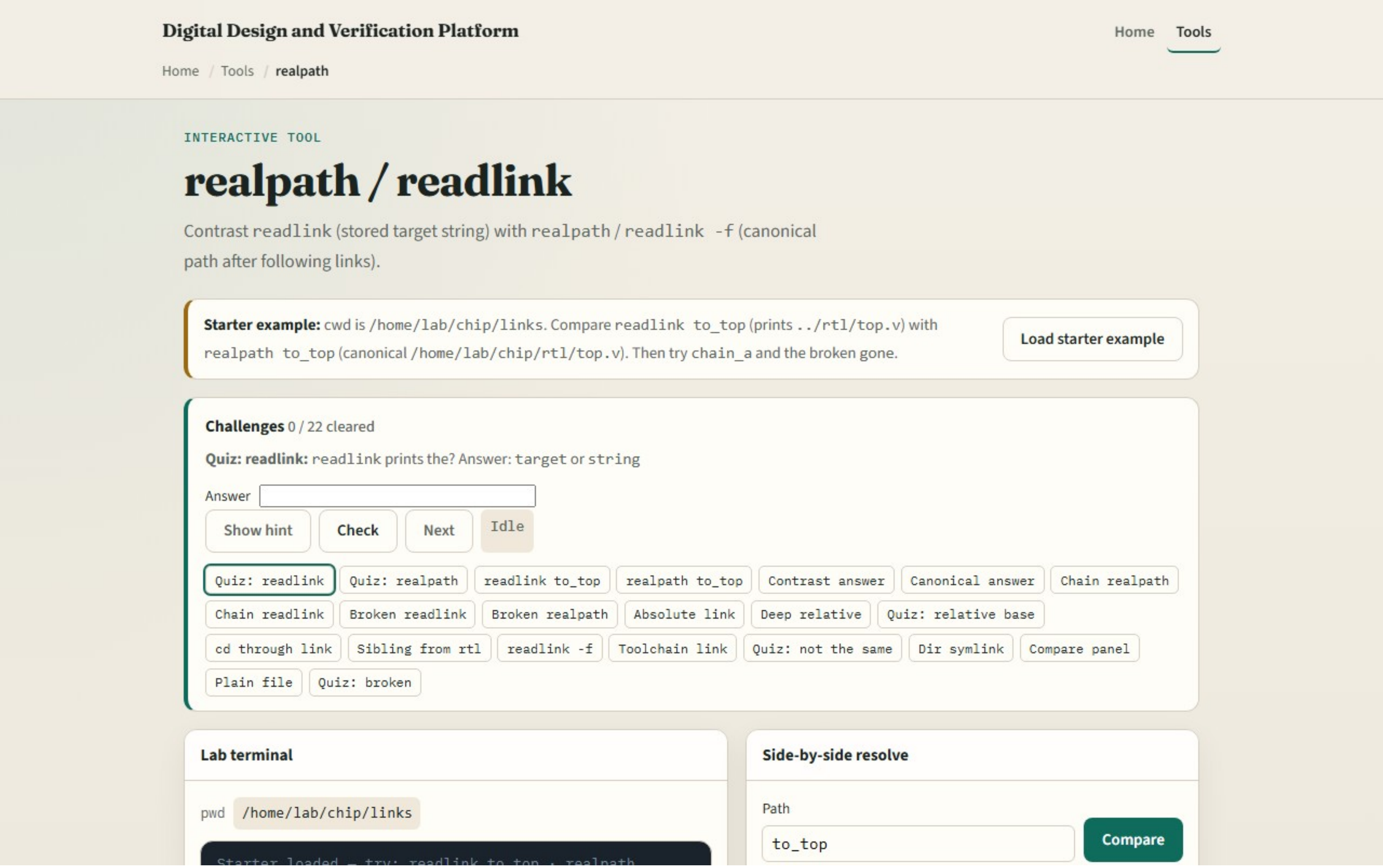
realpath / readlink

Relative paths and symlinks are handy, until a script needs one clear absolute location

Stored target vs canonical path

- Readlink on a symlink prints the stored target string, often a short relative path
- Realpath (and readlink with a force-style flag on Linux) cleans dots and parent segments
- That difference matters in build scripts

Browser lab



realpath resolve lab

Real shell practice

```
wsl — module06-realpath-resolve/examples/realpath

# Track A — real Unix
# real shell session (Track A)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_unix/module06-realpath-resolve/examples/realpath

$ realpath .
/mnt/d/proj/designs/learning/courses/learn_unix/module06-realpath-resolve/examples/realpath

$ realpath a_file.txt
/mnt/d/proj/designs/learning/courses/learn_unix/module06-realpath-resolve/examples/realpath/a_file.txt

$ ls -l link_to_file
lrwxrwxrwx 1 yongfu yongfu 10 Jul 20 03:00 link_to_file -> a_file.txt

$ readlink link_to_file
a_file.txt

$ realpath link_to_file
/mnt/d/proj/designs/learning/courses/learn_unix/module06-realpath-resolve/examples/realpath/a_file.txt

$ readlink -f link_to_file
/mnt/d/proj/designs/learning/courses/learn_unix/module06-realpath-resolve/examples/realpath/a_file.txt
```

Real shell — realpath and readlink

Real shell practice — try these

```
# pwd — where you are before resolving
pwd

# realpath . — absolute canonical path of the current directory
realpath .

# realpath a_file.txt — absolute path of the sample file
realpath a_file.txt

# ls -l link_to_file — see the symlink arrow to the file
ls -l link_to_file

# readlink link_to_file — stored target string (often relative)
readlink link_to_file

# realpath link_to_file — follow the link to a canonical absolute path
```

Pitfalls to watch

- A broken link still has a stored string
- Do not assume every system has the same readlink flags
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, compare readlink’s stored target with realpath’s canonical path
- When you are ready, take the short quiz, then continue to permissions, umask, and PATH